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Class CameraDevice

Namespace: Uralstech.UXR.QuestCamera

Manages a camera device.

```
public sealed class CameraDevice : StatefulResource
```

Inheritance

object ← StatefulResource ← CameraDevice

Inherited Members

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Constructors

CameraDevice(string)

```
public CameraDevice(string cameraId)
```

Parameters

`cameraId` string

Fields

CameraId

The ID of the camera.

```
public readonly string CameraId
```

Field Value

string

NativeProxy

Native callback handler.

```
public readonly CameraDevice.Proxy NativeProxy
```

Field Value

CameraDevice.Proxy

Methods

CreateContinuousPipeline(Resolution, CaptureTemplate, StreamUseCase, GraphicsFormat)

Creates a continuous capture pipeline (session + linked YUV to RGBA converter) for use.

```
public CapturePipeline<ContinuousCaptureSession>?
CreateContinuousPipeline(Resolution resolution, CaptureTemplate template =
CaptureTemplate.Preview, StreamUseCase streamUseCase = StreamUseCase.None,
GraphicsFormat textureFormat = null)
```

Parameters

`resolution` Resolution

`template` CaptureTemplate

`streamUseCase` StreamUseCase

`textureFormat` GraphicsFormat

See YUVConverter(Resolution, ComputeShaderKernel, GraphicsFormat) for default texture format.

Returns

CapturePipeline<ContinuousCaptureSession>

The created pipeline, or null if creation failed.

CreateContinuousSession(Resolution, CaptureTemplate, StreamUseCase)

Creates a continuous capture session for use.

```
public ContinuousCaptureSession CreateContinuousSession(Resolution resolution,
CaptureTemplate template = CaptureTemplate.Preview, StreamUseCase streamUseCase
= StreamUseCase.None)
```

Parameters

`resolution` Resolution

The capture resolution. Must be from SupportedResolutions.

`template` CaptureTemplate

The template to use for the captures.

`streamUseCase` StreamUseCase

The stream use case for this session. Must be from SupportedStreamUseCases or None.

Returns

ContinuousCaptureSession

Returns the session. Check State (inherited by ContinuousCaptureSession) for the state of the session.

Remarks

To shut down and dispose the session, use DisposeAsync() (inherited by ContinuousCapture Session).

Exceptions

ObjectDisposedException

CreateGLESSessionAsync(Resolution, CaptureTemplate, StreamUseCase, GraphicsFormat)

Creates a generic OpenGL-ES based capture session.

```
public ValueTask<GLESCaptureSession> CreateGLESSessionAsync(Resolution resolution,
CaptureTemplate template = CaptureTemplate.Preview, StreamUseCase streamUseCase =
StreamUseCase.None, GraphicsFormat textureFormat = null)
```

Parameters

`resolution` Resolution

The capture resolution. Must be from SupportedResolutions.

`template` CaptureTemplate

The template to use for the captures.

`streamUseCase` StreamUseCase

The stream use case for this session. Must be from SupportedStreamUseCases or None.

`textureFormat` GraphicsFormat

The output texture format for the converted frames. See GLESCaptureSession(Resolution, GraphicsFormat) for default.

Returns

ValueTask<GLESCaptureSession>

Returns the session. Check State (inherited by GLESCaptureSession) for the state of the session.

Remarks

This initializes a native capture session backed by a SurfaceTexture and a GLES conversion job. The returned session must be started manually (e.g., via its run loop or single-run methods) and disposed using DisposeAsync().

Exceptions

ObjectDisposedException

CreateOnDemandPipeline(Resolution, StreamUseCase, GraphicsFormat)

Creates an on-demand capture pipeline (session + linked YUV to RGBA converter) for use.

```
public CapturePipeline<OnDemandCaptureSession>? CreateOnDemandPipeline(Resolution
resolution, StreamUseCase streamUseCase = StreamUseCase.None, GraphicsFormat
textureFormat = null)
```

Parameters

`resolution` Resolution

`streamUseCase` StreamUseCase

`textureFormat` GraphicsFormat

See YUVConverter(Resolution, ComputeShaderKernel, GraphicsFormat) for default texture format.

Returns

CapturePipeline<OnDemandCaptureSession>

The created pipeline, or null if creation failed.

CreateOnDemandSession(Resolution, StreamUseCase)

Creates an on-demand capture session for use.

```
public OnDemandCaptureSession CreateOnDemandSession(Resolution resolution,
StreamUseCase streamUseCase = StreamUseCase.None)
```

Parameters

`resolution` Resolution

The capture resolution. Must be from SupportedResolutions.

`streamUseCase` StreamUseCase

The stream use case for this session. Must be from SupportedStreamUseCases or None.

Returns

OnDemandCaptureSession

Returns the session. Check State (inherited by OnDemandCaptureSession) for the state of the session.

Remarks

To shut down and dispose the session, use DisposeAsync() (inherited by OnDemandCapture Session).

Exceptions

ObjectDisposedException

CreateVulkanContinuousSession(Resolution, CaptureTemplate, StreamUseCase, GraphicsFormat)

Creates a Vulkan-based continuous capture session for use.

```
public VulkanContinuousCaptureSession CreateVulkanContinuousSession(Resolution
resolution, CaptureTemplate template = CaptureTemplate.Preview, StreamUseCase
streamUseCase = StreamUseCase.None, GraphicsFormat textureFormat = null)
```

Parameters

`resolution` Resolution

The capture resolution. Must be from SupportedResolutions.

`template` CaptureTemplate

The template to use for the captures.

`streamUseCase` StreamUseCase

The stream use case for this session. Must be from SupportedStreamUseCases or None.

`textureFormat` GraphicsFormat

The output texture format for the converted frames. See VulkanContinuousCapture Session(Resolution, GraphicsFormat) for default.

Returns

VulkanContinuousCaptureSession

Returns the session. Check State (inherited by VulkanContinuousCaptureSession) for the state of the session.

Remarks

To shut down and dispose the session, use DisposeAsync().

Exceptions

ObjectDisposedException

CreateVulkanOnDemandSession(Resolution, CaptureTemplate, StreamUseCase, GraphicsFormat)

Creates a Vulkan-based on-demand capture session for use.

```
public VulkanOnDemandCaptureSession CreateVulkanOnDemandSession(Resolution
resolution, CaptureTemplate template = CaptureTemplate.Preview, StreamUseCase
streamUseCase = StreamUseCase.None, GraphicsFormat textureFormat = null)
```

Parameters

`resolution` Resolution

The capture resolution. Must be from SupportedResolutions.

`template` CaptureTemplate

The template to use for the captures.

`streamUseCase` StreamUseCase

The stream use case for this session. Must be from SupportedStreamUseCases or None.

`textureFormat` GraphicsFormat

The output texture format for the converted frames. See VulkanOnDemandCapture Session(Resolution, GraphicsFormat) for default.

Returns

VulkanOnDemandCaptureSession

Returns the session. Check State (inherited by VulkanOnDemandCaptureSession) for the state of the session.

Remarks

To shut down and dispose the session, use DisposeAsync() (inherited by VulkanOnDemand CaptureSession).

Exceptions

ObjectDisposedException

DisposeAsync()

Closes the camera (if not already closed) and releases native resources.

```
public ValueTask DisposeAsync()
```

Returns

ValueTask

~CameraDevice()

```
protected ~CameraDevice()
```

ThrowIfDisposed()

```
protected override void ThrowIfDisposed()
```

Events

OnDeviceClosed

Invoked when the camera is closed, along with the camera ID.

```
public event Action<string>? OnDeviceClosed
```

Event Type

Action<string>

OnDeviceDisconnected

Invoked when the camera is disconnected, along with the camera ID.

```
public event Action<string>? OnDeviceDisconnected
```

Event Type

Action<string>

OnDeviceErred

Invoked when the camera encounters an error, along with the camera ID.

```
public event Action<string, CameraDevice.ErrorCode>? OnDeviceErred
```

Event Type

Action<string, CameraDevice.ErrorCode>

OnDeviceOpened

Invoked when the camera is opened, along with the camera ID.

```
public event Action<string>? OnDeviceOpened
```

Event Type

Action<string>

Enum CameraDevice.ErrorCode

Namespace: Uralstech.UXR.QuestCamera

Error codes from the native plugin.

```
public enum CameraDevice.ErrorCode
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

`CameraAccessError = 1001`

The camera is disabled by device policy, has been disconnected, or is being used by a higher-priority camera API client.

`CameraDeviceError = 4`

The camera device has encountered a fatal error.

`CameraDisabled = 3`

The camera device could not be opened due to a device policy.

`CameraInUse = 1`

The camera device is in use already.

`CameraServiceError = 5`

The camera service has encountered a fatal error.

`IllegalArgumentError = 1000`

cameraId was null, or the cameraId does not match any currently or previously available camera device.

`MaxCamerasInUse = 2`

The camera device could not be opened because there are too many other open camera devices.

`SecurityError = 1002`

The application does not have permission to access the camera.

`Unknown = 0`

Class CameraDevice.Proxy

Namespace: Uralstech.UXR.QuestCamera

Java proxy to handle native callbacks.

```
public sealed class CameraDevice.Proxy : AndroidJavaProxy
```

Inheritance

object ← CameraDevice.Proxy

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

All event callbacks will be on a Java thread, and are performance sensitive.

Constructors

Proxy()

```
public Proxy()
```

Events

OnClosed

Invoked when the camera is closed.

```
public event Action? OnClosed
```

Event Type

Action

OnDisconnected

Invoked when the camera is disconnected.

```
public event Action? OnDisconnected
```

Event Type

Action

OnErred

Invoked when the camera encounters an error.

```
public event Action<CameraDevice.ErrorCode>? OnErred
```

Event Type

Action<CameraDevice.ErrorCode>

OnOpened

Invoked when the camera is opened.

```
public event Action? OnOpened
```

Event Type

Action

Class CameraInfo

Namespace: Uralstech.UXR.QuestCamera

Information describing a camera device.

```
public sealed class CameraInfo : CameraMetadata
```

Inheritance

object ← CameraMetadata ← CameraInfo

Inherited Members

CameraMetadata.Native , CameraMetadata.KeyProviderClass , CameraMetadata.GetConstant(string) , CameraMetadata.GetKeys() , CameraMetadata.TryGet<T>(string, out T) , CameraMetadata.TryGet<T>(CameraMetadata.Key, out T) , CameraMetadata.Dispose()

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

CameraInfo(string, AndroidJavaObject, Key?, Key?)

```
public CameraInfo(string cameraId, AndroidJavaObject native, CameraMetadata.Key?
metaQuestSourceKey = null, CameraMetadata.Key? metaQuestPositionKey = null)
```

Parameters

`cameraId` string

`native` AndroidJavaObject

`metaQuestSourceKey` CameraMetadata.Key

`metaQuestPositionKey` CameraMetadata.Key

Fields

CameraId

The device ID of the camera.

```
public readonly string CameraId
```

Field Value

string

Eye

The eye the camera is closest to.

```
public readonly CameraInfo.CameraEye Eye
```

Field Value

[CameraInfo](#).[CameraEye](#)

Intrinsics

The intrinsic data for this camera.

```
public readonly CameraInfo.CameraIntrinsics? Intrinsics
```

Field Value

[CameraInfo](#).[CameraIntrinsics](#)

LensPoseRotation

The orientation of the camera relative to the sensor coordinate system, converted from Android sensor rotation space to Unity rotation space.

```
public readonly Quaternion? LensPoseRotation
```

Field Value

Quaternion?

LensPoseTranslation

The position of the camera's optical center, converted from Android sensor space to Unity space.

```
public readonly Vector3? LensPoseTranslation
```

Field Value

Vector3?

Source

The source of the camera feed.

```
public readonly CameraInfo.CameraSource Source
```

Field Value

CameraInfo.CameraSource

SupportedResolutions

The resolutions supported by this camera.

```
public readonly Resolution[] SupportedResolutions
```

Field Value

Resolution[]

SupportedStreamUseCases

The stream use cases supported by this camera, if any.

```
public readonly StreamUseCase[] SupportedStreamUseCases
```

Field Value

StreamUseCase[]

Remarks

Guaranteed to an empty array on devices < API Level 33.

Methods

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

Equals(CameraInfo?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(CameraInfo? other)
```

Parameters

`other` CameraInfo

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

GetAvailableCaptureRequestKeys()

Returns the keys supported by this CameraDevice for querying with a CaptureRequest.

```
public CameraMetadata.Key[] GetAvailableCaptureRequestKeys()
```

Returns

Key[]

GetAvailableCaptureResultKeys()

Returns the keys supported by this CameraDevice for querying with a CaptureResult.

```
public CameraMetadata.Key[] GetAvailableCaptureResultKeys()
```

Returns

Key[]

GetAvailableSessionCharacteristicsKeys()

Returns the keys for this CameraDevice whose values are capture session specific.

```
public CameraMetadata.Key[] GetAvailableSessionCharacteristicsKeys()
```

Returns

Key[]

Remarks

Returns an empty array on devices < API Level 35.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

GetKeysNeedingPermission()

Returns a subset of the keys returned by GetKeys() with all keys that require camera clients to obtain the Manifest.permission.CAMERA permission.

```
public CameraMetadata.Key[] GetKeysNeedingPermission()
```

Returns

Key[]

ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

Returns

string

A string that represents the current object.

Operators

operator ==(CameraInfo?, CameraInfo?)

```
public static bool operator ==(CameraInfo? a, CameraInfo? b)
```

Parameters

a [CameraInfo]

b [CameraInfo]

Returns

bool

operator !=(CameraInfo?, CameraInfo?)

```
public static bool operator !=(CameraInfo? a, CameraInfo? b)
```

Parameters

a [CameraInfo]

b [CameraInfo]

Returns

bool

Enum CameraInfo.CameraEye

Namespace: Uralstech.UXR.QuestCamera

The camera eye.

```
public enum CameraInfo.CameraEye
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

`Left = 0`

`Right = 1`

`Unknown = -1`

Class CameraInfo.CameraIntrinsics

Namespace: Uralstech.UXR.QuestCamera

Defines the camera's intrinsic properties. All values are in pixels.

```
public sealed record CameraInfo.CameraIntrinsics
```

Inheritance

object ← CameraInfo.CameraIntrinsics

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

CameraIntrinsics(Vector2, Vector2, Vector2, float)

Defines the camera's intrinsic properties. All values are in pixels.

```
public CameraIntrinsics(Vector2 Resolution, Vector2 FocalLength, Vector2
PrincipalPoint, float Skew)
```

Parameters

`Resolution` Vector2

Resolution in pixels.

`FocalLength` Vector2

Focal length in pixels.

`PrincipalPoint` Vector2

Principal point in pixels from the image's top-left corner.

`Skew` float

Skew coefficient for axis misalignment.

Properties

FocalLength

Focal length in pixels.

```
public Vector2 FocalLength { get; init; }
```

Property Value

Vector2

PrincipalPoint

Principal point in pixels from the image's top-left corner.

```
public Vector2 PrincipalPoint { get; init; }
```

Property Value

Vector2

Resolution

Resolution in pixels.

```
public Vector2 Resolution { get; init; }
```

Property Value

Vector2

Skew

Skew coefficient for axis misalignment.

```
public float Skew { get; init; }
```

Property Value

float

Enum CameraInfo.CameraSource

Namespace: Uralstech.UXR.QuestCamera

The source of the camera feed.

```
public enum CameraInfo.CameraSource
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

`PassthroughRGB = 0`

`Unknown = -1`

Class CameraMetadata

Namespace: Uralstech.UXR.QuestCamera

Represents a map of camera device, capture request or capture result metadata.

```
public abstract class CameraMetadata
```

Inheritance

object ← CameraMetadata

Derived

CameraInfo, CaptureRequest, CaptureRequest.Builder, CaptureResult

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

CameraMetadata(AndroidJavaObject, string)

```
public CameraMetadata(AndroidJavaObject native, string className)
```

Parameters

`native` AndroidJavaObject

`className` string

Fields

KeyProviderClass

The Java/Kotlin class which provides the keys and constants for this instance.

```
public readonly AndroidJavaClass KeyProviderClass
```

Field Value

AndroidJavaClass

Native

The native object.

```
public readonly AndroidJavaObject Native
```

Field Value

AndroidJavaObject

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

GetConstant(string)

Gets a constant value from android.hardware.camera2.CameraMetadata.

```
public int GetConstant(string name)
```

Parameters

`name` string

The name of the constant field.

Returns

int

GetKeys()

Returns the keys contained in this map.

```
public CameraMetadata.Key[] GetKeys()
```

Returns

Key[]

Remarks

The caller is responsible for the disposal of all returned CameraMetadata.Key objects.

GetKeysFromListNative(string)

```
protected CameraMetadata.Key[] GetKeysFromListNative(string methodName)
```

Parameters

`methodName` string

Returns

Key[]

ThrowIfDisposed()

```
protected void ThrowIfDisposed()
```

TryGet<T>(string, out T)

Tries to get a value from this map.

```
public bool TryGet<T>(string keyName, out T value)
```

Parameters

`keyName` string

The name of the key.

`value` T

The result.

Returns

bool

true if the key exists and was converted successfully into a managed type, false otherwise.

Type Parameters

`T`

The type of the value.

Remarks

Supports converting to the types supported by ToManaged<T>(AndroidJavaObject) + AndroidJavaObject.

Exceptions

KeyNotFoundException

Thrown if the key is not defined in KeyProviderClass.

TryGet<T>(Key, out T)

Tries to get a value from this map.

```
public bool TryGet<T>(CameraMetadata.Key key, out T value)
```

Parameters

`key` CameraMetadata.Key

The key.

`value` T

The result.

Returns

bool

true if the key exists and was converted successfully into a managed type, false otherwise.

Type Parameters

`T`

The type of the value.

Remarks

Supports converting to the types supported by ToManaged<T>(AndroidJavaObject) + AndroidJavaObject.

Class CameraMetadata.FloatRange

Namespace: Uralstech.UXR.QuestCamera

A range of two float values.

```
public sealed record CameraMetadata.FloatRange
```

Inheritance

object ← CameraMetadata.FloatRange

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

FloatRange(float, float)

A range of two float values.

```
public FloatRange(float Lower, float Upper)
```

Parameters

`Lower` float

The lower endpoint.

`Upper` float

The upper endpoint.

Properties

Lower

```
public float Lower { get; init; }
```

Property Value

float

Upper

```
public float Upper { get; init; }
```

Property Value

float

Class CameraMetadata.IntRange

Namespace: [Uralstech](#).[UXR](#).[QuestCamera](#)

A range of two integer values.

```
public sealed record CameraMetadata.IntRange
```

Inheritance

object ← CameraMetadata.IntRange

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

IntRange(int, int)

A range of two integer values.

```
public IntRange(int Lower, int Upper)
```

Parameters

`Lower` int

The lower endpoint.

`Upper` int

The upper endpoint.

Properties

Lower

The lower endpoint.

```
public int Lower { get; init; }
```

Property Value

int

Upper

The upper endpoint.

```
public int Upper { get; init; }
```

Property Value

int

Class CameraMetadata.Key

Namespace: Uralstech.UXR.QuestCamera

Light wrapper for CaptureRequest.Key, CameraCharacteristics.Key, etc.

```
public sealed class CameraMetadata.Key
```

Inheritance

object ← CameraMetadata.Key

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

DO NOT mix or compare keys from different sources!

Constructors

Key(AndroidJavaObject)

```
public Key(AndroidJavaObject native)
```

Parameters

`native` AndroidJavaObject

Fields

Name

The name of the key.

```
public readonly string Name
```

Field Value

string

Native

The native object.

```
public readonly AndroidJavaObject Native
```

Field Value

AndroidJavaObject

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

Equals(Key?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(CameraMetadata.Key? other)
```

Parameters

`other` CameraMetadata.Key

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

Returns

string

A string that represents the current object.

Operators

operator ==(Key?, Key?)

```
public static bool operator ==(CameraMetadata.Key? a, CameraMetadata.Key? b)
```

Parameters

a [CameraMetadata](#).[Key](#)

b [CameraMetadata](#).[Key](#)

Returns

bool

operator !=(Key?, Key?)

```
public static bool operator !=(CameraMetadata.Key? a, CameraMetadata.Key? b)
```

Parameters

a [CameraMetadata](#).[Key](#)

b [CameraMetadata](#).[Key](#)

Returns

bool

Class CameraMetadata.LongRange

Namespace: Uralstech.UXR.QuestCamera

A range of two long values.

```
public sealed record CameraMetadata.LongRange
```

Inheritance

object ← CameraMetadata.LongRange

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

LongRange(long, long)

A range of two long values.

```
public LongRange(long Lower, long Upper)
```

Parameters

`Lower` long

The lower endpoint.

`Upper` long

The upper endpoint.

Properties

Lower

```
public long Lower { get; init; }
```

Property Value

long

Upper

```
public long Upper { get; init; }
```

Property Value

long

Class CaptureFailure

Namespace: Uralstech.UXR.QuestCamera

A report of failed capture for a single image capture from the image sensor.

```
public sealed class CaptureFailure
```

Inheritance

object ← CaptureFailure

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

CaptureFailure(AndroidJavaObject)

```
public CaptureFailure(AndroidJavaObject native)
```

Parameters

`native` AndroidJavaObject

Fields

FrameNumber

The frame number associated with this failed capture.

```
public readonly long FrameNumber
```

Field Value

long

Native

The native object.

```
public readonly AndroidJavaObject Native
```

Field Value

AndroidJavaObject

Reason

The reason for the capture result being dropped.

```
public readonly CaptureFailure.ErrorCode Reason
```

Field Value

CaptureFailure.ErrorCode

SequenceId

The sequence ID of the capture request originally returned in OnRequestSetWithId.

```
public readonly int SequenceId
```

Field Value

int

WasImageCaptured

If the image was captured from the camera.

```
public readonly bool WasImageCaptured
```

Field Value

bool

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

GetRequest()

Returns the capture request associated with this failure.

```
public CaptureRequest GetRequest()
```

Returns

CaptureRequest

Enum CaptureFailure.ErrorCode

Namespace: Uralstech.UXR.QuestCamera

Error codes from Camera2.

```
public enum CaptureFailure.ErrorCode
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

`FlushedError = 1`

The capture has failed due to an abort call from the application.

`FrameworkError = 0`

The CaptureResult has been dropped this frame only due to an error in the framework.

Class CapturePipeline<T>

Namespace: Uralstech.UXR.QuestCamera

A wrapper for a complete capture-conversion pipeline.

```
public sealed class CapturePipeline<T> where T : ContinuousCaptureSession
```

Type Parameters

`T`

The capture session type.

Inheritance

object ← CapturePipeline<T>

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

CapturePipeline(T, YUVConverter)

```
public CapturePipeline(T session, YUVConverter converter)
```

Parameters

`session` T

`converter` YUVConverter

Fields

Converter

The YUV to RGBA texture converter.

```
public readonly YUVConverter Converter
```

Field Value

YUVConverter

Session

The capture session.

```
public readonly T Session
```

Field Value

T

Methods

DisposeAsync()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources asynchronously.

```
public ValueTask DisposeAsync()
```

Returns

ValueTask

A task that represents the asynchronous dispose operation.

Class CaptureRequest

Namespace: Uralstech.UXR.QuestCamera

An immutable package of settings and outputs needed to capture a single image from the camera device.

```
public sealed class CaptureRequest : CameraMetadata
```

Inheritance

object ← CameraMetadata ← CaptureRequest

Inherited Members

CameraMetadata.Native , CameraMetadata.KeyProviderClass , CameraMetadata.GetConstant(string) , CameraMetadata.GetKeys() , CameraMetadata.TryGet<T>(string, out T) , CameraMetadata.TryGet<T>(CameraMetadata.Key, out T) , CameraMetadata.Dispose()

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

CaptureRequest(AndroidJavaObject)

```
public CaptureRequest(AndroidJavaObject native)
```

Parameters

`native` AndroidJavaObject

Methods

GetTag<T>()

Gets the tag of this request previously set using SetTag<T>(T).

```
public T? GetTag<T>()
```

Returns

T

Type Parameters

`T`

Remarks

Supports the types supported by ToManaged<T>(AndroidJavaObject) + AndroidJavaObject.

Class CaptureRequest.Builder

Namespace: Uralstech.UXR.QuestCamera

Builder for a capture request.

```
public sealed class CaptureRequest.Builder : CameraMetadata
```

Inheritance

object ← CameraMetadata ← CaptureRequest.Builder

Inherited Members

CameraMetadata.Native , CameraMetadata.KeyProviderClass , CameraMetadata.GetConstant(string) , CameraMetadata.GetKeys() , CameraMetadata.TryGet<T>(string, out T) , CameraMetadata.TryGet<T>(CameraMetadata.Key, out T) , CameraMetadata.Dispose()

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

Builder(AndroidJavaObject)

```
public Builder(AndroidJavaObject native)
```

Parameters

`native` AndroidJavaObject

Methods

SetTag<T>(T)

Set a tag for this request.

```
public void SetTag<T>(T tag) where T : notnull
```

Parameters

`tag` T

An arbitrary Object to store with this request.

Type Parameters

`T`

The type of the tag.

Remarks

Supports the types supported by [ToJava<T>(T)] + AndroidJavaObject.

Set<T>(string, T)

Set a capture request field to a value.

```
public void Set<T>(string keyName, T value) where T : notnull
```

Parameters

`keyName` string

The name of the key.

`value` T

The value to set.

Type Parameters

`T`

The type of the value.

Remarks

Supports the types supported by ToJava<T>(T) + AndroidJavaObject.

Exceptions

KeyNotFoundException

Thrown if the key is not defined in KeyProviderClass.

Set<T>(Key, T)

Set a capture request field to a value.

```
public void Set<T>(CameraMetadata.Key key, T value) where T : notnull
```

Parameters

`key` CameraMetadata.Key

The key.

`value` T

The value to set.

Type Parameters

`T`

The type of the value.

Remarks

Supports the types supported by ToJava<T>(T) + AndroidJavaObject.

Class CaptureResult

Namespace: Uralstech.UXR.QuestCamera

The subset of the results of a single image capture from the image sensor.

```
public class CaptureResult : CameraMetadata
```

Inheritance

object ← CameraMetadata ← CaptureResult

Derived

TotalCaptureResult

Inherited Members

CameraMetadata.Native , CameraMetadata.KeyProviderClass , CameraMetadata.GetConstant(string) , CameraMetadata.GetKeys() , CameraMetadata.GetKeysFromListNative(string) , CameraMetadata.TryGet<T>(string, out T) , CameraMetadata.TryGet<T>(CameraMetadata.Key, out T) , CameraMetadata.Dispose() , CameraMetadata.ThrowIfDisposed()

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

CaptureResult(AndroidJavaObject)

```
public CaptureResult(AndroidJavaObject native)
```

Parameters

`native` AndroidJavaObject

Methods

GetCameraId()

Returns the camera ID of the camera that produced this capture result.

```
public string GetCameraId()
```

Returns

string

GetFrameNumber()

Returns the frame number associated with this result.

```
public long GetFrameNumber()
```

Returns

long

GetRequest()

Returns the capture request associated with this result.

```
public CaptureRequest GetRequest()
```

Returns

CaptureRequest

GetSequenceId()

Returns the sequence ID of the capture request originally returned in OnRequestSetWithId.

```
public int GetSequenceId()
```

Returns

int

Class CaptureSessionBase<TProxy>

Namespace: Uralstech.UXR.QuestCamera

Base class for all capture session types.

```
public abstract class CaptureSessionBase<TProxy> : StatefulResource where TProxy :
CaptureSessionBase<TProxy>.ProxyBase
```

Type Parameters

`TProxy`

The proxy type.

Inheritance

object ← StatefulResource ← CaptureSessionBase<TProxy>

Derived

ContinuousCaptureSession, GLESCaptureSession, VulkanContinuousCaptureSession

Inherited Members

StatefulResource.State , StatefulResource.WaitForInitialization() ,
StatefulResource.WaitForInitialization(TimeSpan, Action, WaitTimeoutMode) ,
StatefulResource.WaitForInitializationAsync(CancellationToken)

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

CaptureSessionBase(TProxy, AndroidJavaObject)

```
protected CaptureSessionBase(TProxy proxy, AndroidJavaObject native)
```

Parameters

`proxy` TProxy

`native` AndroidJavaObject

Fields

NativeProxy

Native callback handler.

```
public readonly TProxy NativeProxy
```

Field Value

TProxy

_disposed

```
protected bool _disposed
```

Field Value

bool

_native

```
protected readonly AndroidJavaObject _native
```

Field Value

AndroidJavaObject

Methods

CloseWork()

```
protected ValueTask CloseWork()
```

Returns

ValueTask

DisposeAsync()

Closes the session (if not already closed) and releases native resources.

```
public virtual ValueTask DisposeAsync()
```

Returns

ValueTask

~CaptureSessionBase()

```
protected ~CaptureSessionBase()
```

MakeProxy<T>(out T)

```
protected static T MakeProxy<T>(out T proxy) where T :
CaptureSessionBase<TProxy>.ProxyBase, new()
```

Parameters

`proxy` T

Returns

T

Type Parameters

T

ThrowIfDisposed()

```
protected override void ThrowIfDisposed()
```

TryAbortCaptures(out ErrorCode)

Discard all captures currently pending and in-progress as fast as possible.

```
public bool TryAbortCaptures(out CaptureSessionBase<TProxy>.ErrorCode errorCode)
```

Parameters

`errorCode` CaptureSessionBase<TProxy>.ErrorCode

Error code if the operation was unsuccessful.

Returns

bool

true if successful; false otherwise.

Remarks

This makes the session unusable for the future, so call DisposeAsync() afterwards.

Exceptions

ObjectDisposedException

Events

OnSessionClosed

Called when the session is closed.

```
public event Action? OnSessionClosed
```

Event Type

Action

OnSessionConfigurationFailed

Called when the session could not be configured.

```
public event Action<CaptureSessionBase<TProxy>.ErrorCode>?
OnSessionConfigurationFailed
```

Event Type

Action<CaptureSessionBase<TProxy>.ErrorCode>

OnSessionConfigured

Called when the session has been configured.

```
public event Action? OnSessionConfigured
```

Event Type

Action

OnSessionRequestFailed

Called when the session request could not be set.

```
public event Action<CaptureSessionBase<TProxy>.ErrorCode>? OnSessionRequestFailed
```

Event Type

Action<CaptureSessionBase<TProxy>.ErrorCode>

OnSessionRequestSet

Called when the session request has been set.

```
public event Action? OnSessionRequestSet
```

Event Type

Action

OnSessionRequestSetWithId

Same as OnSessionRequestSet, but includes the sequence ID of the capture request.

```
public event Action<int>? OnSessionRequestSetWithId
```

Event Type

Action<int>

Enum CaptureSessionBase<TProxy>.ErrorCode

Namespace: Uralstech.UXR.QuestCamera

Error codes from the native plugin.

```
public enum CaptureSessionBase<TProxy>.ErrorCode
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

`CameraAccessError = 1001`

In case the camera device is no longer connected or has encountered a fatal error.

`ConfigurationFailed = 2000`

The session could not be configured.

`IllegalArgumentError = 1000`

In case the session configuration is invalid, if the captureTemplate is not supported by this device, or other illegal arguments were passed.

`IllegalStateError = 1003`

In case the camera device or session has been closed.

`NativeJobBindingFailed = 3000`

The Kotlin session could not bind to its C++ job.

`OutOfResourcesError = 1004`

If the surface or surfaceTexture for the session could not be created due to an out-of-resources error.

Delegate CaptureSessionBase<TProxy>.ModifyRequestBuilderCallback

Namespace: Uralstech.UXR.QuestCamera

A callback for modification of all CaptureRequest builders created for the session.

```
public delegate void
CaptureSessionBase<TProxy>.ModifyRequestBuilderCallback(CaptureRequest.Builder
builder, bool isRepeatingRequest)
```

Parameters

`builder` CaptureRequest.Builder

The builder object. DO NOT dispose this, or persist a reference to it beyond the callback.

`isRepeatingRequest` bool

If this builder is for a repeating or on-demand capture request.

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Class CaptureSessionBase<TProxy>.ProxyBase

Namespace: Uralstech.UXR.QuestCamera

Java proxy to handle native callbacks.

```
public abstract class CaptureSessionBase<TProxy>.ProxyBase : AndroidJavaProxy
```

Inheritance

object ← CaptureSessionBase<TProxy>.ProxyBase

Derived

ContinuousCaptureSession.Proxy, GLESCaptureSession.Proxy, VulkanContinuousCaptureSession.Proxy

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

All event callbacks will be on a Java thread, and are performance sensitive.

Constructors

ProxyBase()

```
public ProxyBase()
```

ProxyBase(string)

```
protected ProxyBase(string className)
```

Parameters

`className` string

Events

ModifyRequestBuilder

A callback for modification of all CaptureRequest builders created for the session.

```
public event CaptureSessionBase<TProxy>.ModifyRequestBuilderCallback?
ModifyRequestBuilder
```

Event Type

CaptureSessionBase<TProxy>.ModifyRequestBuilderCallback

Remarks

You should generally avoid configuring the request if `isRepeatingRequest` is true for on-demand sessions.

OnCaptureCompleted

Called when an image capture has fully completed and all the result metadata is available.

```
public event Action<CaptureRequest, TotalCaptureResult>? OnCaptureCompleted
```

Event Type

Action<CaptureRequest, TotalCaptureResult>

Remarks

Override ShouldRegisterCaptureEvents for capture requests that should invoke this event. DO NOT persist references beyond the callback.

OnCaptureFailed

Called instead of OnCaptureCompleted when the camera device failed to produce a CaptureResult for the request.

```
public event Action<CaptureRequest, CaptureFailure>? OnCaptureFailed
```

Event Type

Action<CaptureRequest, CaptureFailure>

Remarks

Override ShouldRegisterCaptureEvents for capture requests that should invoke this event. DO NOT persist references beyond the callback.

OnCaptureSequenceAborted

Called when a capture sequence aborts before any CaptureResult or CaptureFailure for it have been returned via this listener.

```
public event Action<int>? OnCaptureSequenceAborted
```

Event Type

Action<int>

Remarks

Override ShouldRegisterCaptureEvents for capture requests that should invoke this event. DO NOT persist references beyond the callback.

OnCaptureSequenceCompleted

Called when a capture sequence finishes and all CaptureResult or CaptureFailure for it have been returned via this listener.

```
public event Action<int, long>? OnCaptureSequenceCompleted
```

Event Type

Action<int, long>

Remarks

Override ShouldRegisterCaptureEvents for capture requests that should invoke this event. DO NOT persist references beyond the callback.

OnClosed

Called when the session is closed.

```
public event Action? OnClosed
```

Event Type

Action

OnConfigureFailed

Called when the session could not be configured.

```
public event Action<CaptureSessionBase<TProxy>.ErrorCode>? OnConfigureFailed
```

Event Type

Action<CaptureSessionBase<TProxy>.ErrorCode>

OnConfigured

Called when the session has been configured.

```
public event Action? OnConfigured
```

Event Type

Action

OnRequestFailed

Called when the session request could not be set.

```
public event Action<CaptureSessionBase<TProxy>.ErrorCode>? OnRequestFailed
```

Event Type

Action<CaptureSessionBase<TProxy>.ErrorCode>

OnRequestSet

Called when the session request has been set.

```
public event Action? OnRequestSet
```

Event Type

Action

OnRequestSetWithId

Same as OnRequestSet, but includes the sequence ID of the capture request.

```
public event Action<int>? OnRequestSetWithId
```

Event Type

Action<int>

ShouldRegisterCaptureEvents

A callback for configuring if capture-specific events should be registered for a capture request. Defaults to false.

```
public event CaptureSessionBase<TProxy>.ShouldRegisterCaptureEventsCallback?
ShouldRegisterCaptureEvents
```

Event Type

CaptureSessionBase<TProxy>.ShouldRegisterCaptureEventsCallback

Remarks

Please only register one listener to this event for each session.

Delegate CaptureSessionBase<TProxy>.ShouldRegisterCaptureEventsCallback

Namespace: Uralstech.UXR.QuestCamera

A callback for configuring if capture-specific events should be registered for a capture request. Defaults to false.

```
public delegate bool
CaptureSessionBase<TProxy>.ShouldRegisterCaptureEventsCallback(CaptureRequest
request, bool isRepeatingRequest)
```

Parameters

`request` CaptureRequest

The request for which events should be registered.

`isRepeatingRequest` bool

If this is for a repeating or on-demand capture request.

Returns

bool

A callback for configuring if capture-specific events should be registered for a capture request. Defaults to false.

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

Please only register one listener to this event for each session.

Enum CaptureTemplate

Namespace: Uralstech.UXR.QuestCamera

Capture template to use when recording.

```
public enum CaptureTemplate
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

`Preview = 1`

Creates a request suitable for a camera preview window.

`Record = 3`

Creates a request suitable for video recording.

`StillCapture = 2`

Creates a request suitable for still image capture.

`VideoSnapshot = 4`

Creates a request suitable for still image capture while recording video.

Class ComputeShaderKernel

Namespace: Uralstech.UXR.QuestCamera

Utility class to contain ComputeShader kernel info.

```
[Serializable]
public sealed class ComputeShaderKernel
```

Inheritance

object ← ComputeShaderKernel

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

ComputeShaderKernel()

```
public ComputeShaderKernel()
```

ComputeShaderKernel(ComputeShader, string)

```
public ComputeShaderKernel(ComputeShader shader, string kernel = "CSMain")
```

Parameters

`shader` ComputeShader

`kernel` string

Properties

Index

```
public int Index { get; }
```

Property Value

int

Name

```
public string Name { get; }
```

Property Value

string

Shader

```
public ComputeShader Shader { get; }
```

Property Value

ComputeShader

ThreadGroupSizes

```
public Vector3Int ThreadGroupSizes { get; }
```

Property Value

Vector3Int

Methods

Validate()

```
public bool Validate()
```

Returns

bool

Class ContinuousCaptureSession

Namespace: Uralstech.UXR.QuestCamera

Manages a camera capture session with a continuous/repeating request.

```
public class ContinuousCaptureSession :
CaptureSessionBase<ContinuousCaptureSession.Proxy>
```

Inheritance

object ← StatefulResource ← CaptureSessionBase<ContinuousCaptureSession.Proxy> ← ContinuousCaptureSession

Derived

OnDemandCaptureSession

Inherited Members

CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionConfigured , CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionConfigurationFailed , CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionRequestSet , CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionRequestSetWithId , CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionRequestFailed , CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionClosed , CaptureSessionBase<ContinuousCaptureSession.Proxy>.NativeProxy , CaptureSessionBase<ContinuousCaptureSession.Proxy>._native , CaptureSessionBase<ContinuousCaptureSession.Proxy>._disposed , CaptureSessionBase<ContinuousCaptureSession.Proxy>.MakeProxy<T>(out T) , CaptureSessionBase<ContinuousCaptureSession.Proxy>.TryAbortCaptures(out CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode) , CaptureSessionBase<ContinuousCaptureSession.Proxy>.DisposeAsync() , CaptureSessionBase<ContinuousCaptureSession.Proxy>.CloseWork() , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ThrowIfDisposed() , StatefulResource.State , StatefulResource.WaitForInitialization() , StatefulResource.WaitForInitialization(TimeSpan, Action, WaitTimeoutMode) , StatefulResource.WaitForInitializationAsync(CancellationToken)

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

This does not do any image processing/conversion on its own, but classes like [YUVConverter] can register callbacks to [OnFrameReady] to do their own processing of the raw image YUV 4:2:0, BT.601 Full Range data.

Constructors

ContinuousCaptureSession(Resolution)

```
public ContinuousCaptureSession(Resolution resolution)
```

Parameters

`resolution` Resolution

ContinuousCaptureSession(Resolution, string)

```
protected ContinuousCaptureSession(Resolution resolution, string className)
```

Parameters

`resolution` Resolution

`className` string

Delegate ContinuousCaptureSession.OnFrameReadyCallback

Namespace: Uralstech.UXR.QuestCamera

Processes a frame received from the native capture session.

```
public delegate void ContinuousCaptureSession.OnFrameReadyCallback(nint yBuffer,
long yBufferSize, nint uBuffer, nint vBuffer, long uvBufferSize, int yRowStride, int
uvRowStride, int uvPixelStride, long timestamp)
```

Parameters

`yBuffer` nint

The pointer to this frame's Y (luminance) data.

`yBufferSize` long

The size of the Y buffer in bytes.

`uBuffer` nint

The pointer to this frame's U (color) data.

`vBuffer` nint

The pointer to this frame's V (color) data.

`uvBufferSize` long

The size of the U and V buffers in bytes.

`yRowStride` int

The size of each row of the image in `yBuffer` in bytes.

`uvRowStride` int

The size of each row of the image in `uBuffer` and `vBuffer` in bytes.

`uvPixelStride` int

The size of a pixel in a row of the image in `uBuffer` and `vBuffer` in bytes.

`timestamp` long

The timestamp the frame was captured at in nanoseconds.

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Class ContinuousCaptureSession.Proxy

Namespace: Uralstech.UXR.QuestCamera

Java proxy to handle native callbacks.

```
public sealed class ContinuousCaptureSession.Proxy :
CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase
```

Inheritance

object ← CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase ← ContinuousCaptureSession.Proxy

Inherited Members

CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.ModifyRequestBuilder , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.ShouldRegisterCaptureEvents , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnConfigured , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnConfigureFailed , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnRequestSet , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnRequestSetWithId , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnRequestFailed , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureCompleted , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureFailed , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceCompleted , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceAborted , CaptureSessionBase<ContinuousCaptureSession.Proxy>.ProxyBase.OnClosed

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

All event callbacks will be on a Java thread, and are performance sensitive.

Constructors

Proxy()

```
public Proxy()
```

Methods

Invoke(string, nint)

```
public override nint Invoke(string methodName, nint javaArgs)
```

Parameters

`methodName` string

`javaArgs` nint

Returns

nint

Events

OnFrameReady

Processes a frame received from the native capture session.

```
public event ContinuousCaptureSession.OnFrameReadyCallback? OnFrameReady
```

Event Type

[ContinuousCaptureSession](#).[OnFrameReadyCallback](#)

Remarks

See [ContinuousCaptureSession.OnFrameReadyCallback](#) for parameters.

Class JNIExtensions

Namespace: Uralstech.UXR.QuestCamera

QOL extensions for the JNI.

```
public static class JNIExtensions
```

Inheritance

object ← JNIExtensions

Remarks

This class is public to allow package users to reuse these extensions if useful. However, it should not be considered stable and is not part of the supported public API. It exists solely as an internal utility and may change or be removed at any time.

Methods

JavaListAsManagedArray<T>(AndroidJavaObject)

Converts a java.util.List to a managed array.

```
public static T[] JavaListAsManagedArray<T>(this AndroidJavaObject current)
```

Parameters

`current` AndroidJavaObject

Returns

T[]

The converted array.

Type Parameters

`T`

The type of the elements.

Remarks

WARNING: This method **DOES NOT** verify if the native Java/Kotlin elements can actually be converted to the target type.

Supports converting elements to the types supported by [ToManaged<T>(AndroidJavaObject)]() + AndroidJavaObject.

ToJava(object, Type)

Converts managed .NET/Unity object to a Java/Kotlin object represented by an AndroidJavaObject.

```
[SuppressMessage("Style", "IDE0046:Convert to conditional expression", Justification
= "Code is neater without conditionals.")]
public static AndroidJavaObject ToJava(this object current, Type target)
```

Parameters

`current` object

`target` Type

The type of the managed object.

Returns

AndroidJavaObject

The converted Java object.

Remarks

Basic supported types:

- bool
- char
- sbyte
- short
- int

- long
- float
- double
- string

Supported Android types:

Unity Type	Android Type
Resolution	android.util.Size
Rect	android.graphics.RectF
RectInt	android.graphics.Rect
CameraMetadata.IntRange	[android.util.Range<Integer>](android.util.Range)
CameraMetadata.LongRange	android.util.Range<Long>
CameraMetadata.FloatRange	android.util.Range<Float>

Also supports converting arrays of all above types + generic AndroidJavaObject[] and AndroidJavaProxy[].

Exceptions

NotSupportedException

Thrown when the type is unsupported.

ToJava<T>(T)

Converts managed .NET/Unity object to a Java/Kotlin object represented by an AndroidJavaObject.

```
public static AndroidJavaObject ToJava<T>(this T current) where T : notnull
```

Parameters

`current` T

Returns

AndroidJavaObject

The converted Java object.

Type Parameters

T

The type of the managed object.

Remarks

Basic supported types:

- bool
- char
- sbyte
- short
- int
- long
- float
- double
- string

Supported Android types:

Unity Type	Android Type
Resolution	android.util.Size
Rect	android.graphics.RectF
RectInt	android.graphics.Rect
CameraMetadata.IntRange	android.util.Range<Integer>
CameraMetadata.LongRange	android.util.Range<Long>
CameraMetadata.FloatRange	android.util.Range<Float>

Also supports converting arrays of all above types + generic AndroidJavaObject[] and AndroidJavaProxy[].

Exceptions

NotSupportedException

Thrown when the type is unsupported.

ToManaged(AndroidJavaObject, Type)

Converts a Java/Kotlin object represented by an AndroidJavaObject into a managed .NET/Unity type.

```
[SuppressMessage("Style", "IDE0046:Convert to conditional expression", Justification
= "Code is neater without conditionals.")]
public static object ToManaged(this AndroidJavaObject current, Type target)
```

Parameters

`current` AndroidJavaObject

`target` Type

The managed target type.

Returns

object

The converted managed object.

Remarks

WARNING: This method **DOES NOT** verify if the native Java/Kotlin object can actually be converted to the target type.

Basic supported types:

- bool
- char
- sbyte
- short
- int
- long

- float
- double
- string

Supported Android types:

Unity Type	Android Type
Resolution	android.util.Size
Rect	android.graphics.RectF
RectInt	android.graphics.Rect
CameraMetadata.IntRange	android.util.Range<Integer>
CameraMetadata.LongRange	android.util.Range<Long>
CameraMetadata.FloatRange	android.util.Range<Float>

Also supports converting arrays of all above types + generic AndroidJavaObject[].

Exceptions

NotSupportedException

Thrown when the requested target type is unsupported.

ToManaged<T>(AndroidJavaObject)

```
public static T ToManaged<T>(this AndroidJavaObject current)
```

Parameters

`current` AndroidJavaObject

Returns

T

Type Parameters

T

The managed target type.

TryConvertToManaged<T>(AndroidJavaObject, out T?)

Safe version of [ToManaged<T>(AndroidJavaObject)].

```
public static bool TryConvertToManaged<T>(this AndroidJavaObject current, out
T? value)
```

Parameters

current AndroidJavaObject

value T

Returns

bool

Type Parameters

T

UnboxBoolElement(nint, int)

Unboxes a boolean from a native Object array.

```
public static bool UnboxBoolElement(nint args, int index)
```

Parameters

args nint

The native array to take the boolean from.

index int

The index of the boolean object in the native array.

Returns

bool

The unboxed boolean.

UnboxByteBufferElement(nint, int)

Retrieves a ByteBuffer object from a native object array and obtains its direct memory address.

```
public static (nint obj, nint ptr) UnboxByteBufferElement(nint args, int index)
```

Parameters

`args` nint

JNI object array containing the ByteBuffer.

`index` int

Index of the ByteBuffer within the array.

Returns

(nint obj, nint ptr)

A tuple containing the local JNI reference to the ByteBuffer, and the pointer to its direct buffer memory.

UnboxIntElement(nint, int)

Unboxes an integer from a native Object array.

```
public static int UnboxIntElement(nint args, int index)
```

Parameters

`args` nint

The native array to take the integer from.

`index` int

The index of the integer object in the native array.

Returns

int

The unboxed integer.

UnboxLongElement(nint, int)

Unboxes a long from a native Object array.

```
public static long UnboxLongElement(nint args, int index)
```

Parameters

`args` nint

The native array to take the long from.

`index` int

The index of the long object in the native array.

Returns

long

The unboxed long.

UnboxObjectElement(nint, int)

Unboxes an AndroidJavaObject from a native Object array.

```
public static AndroidJavaObject UnboxObjectElement(nint args, int index)
```

Parameters

`args` nint

The native array to take the object from.

`index` int

The index of the object in the native array.

Returns

AndroidJavaObject

The unboxed object.

Class OnDemandCaptureSession

Namespace: Uralstech.UXR.QuestCamera

Manages an on-demand, explicit capture session.

```
public sealed class OnDemandCaptureSession : ContinuousCaptureSession
```

Inheritance

object ← StatefulResource ← CaptureSessionBase<ContinuousCaptureSession.Proxy> ← ContinuousCaptureSession ← OnDemandCaptureSession

Inherited Members

CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionConfigured , CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionConfigurationFailed , CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionRequestSet , CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionRequestSetWithId , CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionRequestFailed , CaptureSessionBase<ContinuousCaptureSession.Proxy>.OnSessionClosed , CaptureSessionBase<ContinuousCaptureSession.Proxy>.NativeProxy , CaptureSessionBase<ContinuousCaptureSession.Proxy>.TryAbortCaptures(out CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode) , CaptureSessionBase<ContinuousCaptureSession.Proxy>.DisposeAsync() , StatefulResource.State , StatefulResource.WaitForInitialization() , StatefulResource.WaitForInitialization(TimeSpan, Action, WaitTimeoutMode) , StatefulResource.WaitForInitializationAsync(CancellationToken)

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

This does not do any image processing/conversion on its own, but classes like YUVConverter can register callbacks to OnFrameReady to do their own processing of the raw image YUV 4:2:0, BT.601 Full Range data.

Constructors

OnDemandCaptureSession(Resolution)

```
public OnDemandCaptureSession(Resolution resolution)
```

Parameters

`resolution` Resolution

Methods

RequestCapture(CaptureTemplate)

Requests a new capture from the session.

```
public OnDemandCaptureSession.RequestStatus RequestCapture(CaptureTemplate template
= CaptureTemplate.StillCapture)
```

Parameters

`template` CaptureTemplate

Returns

OnDemandCaptureSession.RequestStatus

Exceptions

ObjectDisposedException

TryRequestCapture(out ErrorCode, CaptureTemplate)

Requests a new capture from the session.

```
public bool TryRequestCapture(out
CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode errorCode,
CaptureTemplate template = CaptureTemplate.StillCapture)
```

Parameters

`errorCode` CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode

Error code if the operation was unsuccessful.

`template` CaptureTemplate

The capture template to use for the capture.

Returns

bool

true if successful; false otherwise.

Exceptions

ObjectDisposedException

Struct OnDemandCaptureSession.RequestStatus

Namespace: Uralstech.UXR.QuestCamera

Status for on-demand capture requests.

```
public readonly struct OnDemandCaptureSession.RequestStatus
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

RequestStatus(ErrorCode, int, bool)

```
public RequestStatus(CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode
errorCode, int sequenceId, bool success)
```

Parameters

`errorCode` CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode

`sequenceId` int

`success` bool

Fields

ErrorCode

Error code for failures, only valid if Success is false.

```
public readonly CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode
ErrorCode
```

Field Value

CaptureSessionBase<ContinuousCaptureSession.Proxy>.ErrorCode

SequenceId

Sequence ID of capture request, only valid if Success is true.

```
public readonly int SequenceId
```

Field Value

int

Success

If the request was successful.

```
public readonly bool Success
```

Field Value

bool

Enum PCASupport

Namespace: Uralstech.UXR.QuestCamera

```
public enum PCASupport
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

`Supported = 1`

PCA API is supported.

Meta XR Core SDK on Quest 3/3S with Horizon OS v74+.

`Unknown = 0`

Support status cannot be determined.

Returned when the Meta XR Core SDK is not available.

`Unsupported = 2`

PCA API is not supported.

Non-Android device, or Meta XR Core SDK on unsupported devices or OS versions.

Class QuestCameraManager

Namespace: Uralstech.UXR.QuestCamera

Entry point for the native Camera2 plugin.

```
public sealed class QuestCameraManager : DontCreateNewSingleton<QuestCameraManager>
```

Inheritance

object ← QuestCameraManager

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

AvatarCameraPermission

Meta Quest Avatar Camera API permission string.

```
public const string AvatarCameraPermission = "android.permission.CAMERA"
```

Field Value

string

ConversionKernel

The shader and kernel to use for YUV 4:2:0 to RGBA conversion.

```
public ComputeShaderKernel ConversionKernel
```

Field Value

ComputeShaderKernel

HeadsetCameraPermission

Meta Quest Passthrough Camera API permission string.

```
public const string HeadsetCameraPermission = "horizonos.permission.HEADSET_CAMERA"
```

Field Value

string

MetaQuestCameraPositionKeyName

Name of the metadata key for the Quest's camera eye position.

```
public const string MetaQuestCameraPositionKeyName =
"com.meta.extra_metadata.position"
```

Field Value

string

MetaQuestCameraSourceKeyName

Name of the metadata key for the Quest's camera source.

```
public const string MetaQuestCameraSourceKeyName =
"com.meta.extra_metadata.camera_source"
```

Field Value

string

Properties

Cameras

A managed, cached array of available cameras and their characteristics.

```
public IReadOnlyList<CameraInfo> Cameras { get; }
```

Property Value

IReadOnlyList<CameraInfo>

MetaQuestCameraPositionKey

Metadata key for the Quest's camera eye position.

```
public CameraMetadata.Key MetaQuestCameraPositionKey { get; }
```

Property Value

CameraMetadata.Key

MetaQuestCameraSourceKey

Metadata key for the Quest's camera source.

```
public CameraMetadata.Key MetaQuestCameraSourceKey { get; }
```

Property Value

CameraMetadata.Key

Support

Tries to get the runtime's support for the Passthrough Camera Access and Camera2 APIs.

```
public static PCASupport Support { get; }
```

Property Value

PCASupport

Methods

Awake()

```
protected override void Awake()
```

GetDevices()

Gets the IDs and intrinsics of all connected camera devices.

```
public CameraInfo[] GetDevices()
```

Returns

CameraInfo[]

OpenCamera(string)

Opens a camera device for use.

```
public CameraDevice OpenCamera(string cameraId)
```

Parameters

`cameraId` string

The ID of the camera to open.

Returns

CameraDevice

The camera device. Check State (inherited by CameraDevice) for the state of the device.

Remarks

Once you have finished using the camera, close and dispose of it using DisposeAsync().

OpenCamera(CameraInfo)

Opens a camera device for use.

```
public CameraDevice OpenCamera(CameraInfo cameraInfo)
```

Parameters

`cameraInfo` CameraInfo

Returns

CameraDevice

The camera device. Check State (inherited by CameraDevice) for the state of the device.

Remarks

Once you have finished using the camera, close and dispose of it using DisposeAsync().

RefreshDevices()

Refreshes cached camera device information.

```
public void RefreshDevices()
```

TryGetDevice(CameraEye, out CameraInfo?)

Finds a camera device by its corresponding eye.

```
public bool TryGetDevice(CameraInfo.CameraEye eye, out CameraInfo? cameraInfo)
```

Parameters

`eye` CameraInfo.CameraEye

`cameraInfo` CameraInfo

Returns

bool

Enum ResourceState

Namespace: Uralstech.UXR.QuestCamera

State of a Camera2 resource.

```
public enum ResourceState
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

`Initializing = 0`

Resource is initializing.

`Invalid = 2`

Resource failed with an error, was disconnected or is being/was closed normally.

`Valid = 1`

Resource is open and ready.

Class StatefulResource

Namespace: Uralstech.UXR.QuestCamera

```
public abstract class StatefulResource
```

Inheritance

object ← StatefulResource

Derived

CameraDevice, CaptureSessionBase<TProxy>

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Properties

State

The state of this resource.

```
public ResourceState State { get; protected set; }
```

Property Value

ResourceState

Remarks

This does NOT indicate disposal status, and is just an indicator for if you can use this resource.

Methods

ThrowIfDisposed()

```
protected abstract void ThrowIfDisposed()
```

WaitForInitialization()

Waits until the resource opens or errs out.

```
public WaitUntil WaitForInitialization()
```

Returns

WaitUntil

Exceptions

ObjectDisposedException

WaitForInitialization(TimeSpan, Action, WaitTimeoutMode)

Waits until the resource opens or errs out.

```
public WaitUntil WaitForInitialization(TimeSpan timeout, Action onTimeout,
WaitTimeoutMode timeoutMode = null)
```

Parameters

`timeout` TimeSpan

Maximum time to wait.

`onTimeout` Action

The action to perform when the `timeout` is reached.

`timeoutMode` WaitTimeoutMode

Mode in which to measure time to determine `timeout`.

Returns

WaitUntil

Exceptions

ObjectDisposedException

WaitForInitializationAsync(CancellationToken)

Waits until the resource opens or errs out.

```
public Task<bool> WaitForInitializationAsync(CancellationToken token = default)
```

Parameters

`token` CancellationToken

Returns

Task<bool>

true if opened successfully, false otherwise.

Exceptions

ObjectDisposedException

Enum StreamUseCase

Namespace: Uralstech.UXR.QuestCamera

Stream Use Cases are a way to improve the performance of Camera2 capture sessions. They give the hardware device more information to tune parameters, which provides a better camera experience for your specific task.

```
public enum StreamUseCase : long
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

`Default = 0`

Default stream use case.

`None = -1`

Does not set any stream use case. Use this if the device does not support stream use cases or you're unsure of support.

`Preview = 1`

Live stream shown to the user.

`PreviewVideoStill = 4`

One single stream used for combined purposes of preview, video, and still capture.

`StillCapture = 2`

Still photo capture.

`VideoCall = 5`

Long-running video call optimized for both power efficiency and video quality.

`VideoRecord = 3`

Recording video clips.

Class TotalCaptureResult

Namespace: Uralstech.UXR.QuestCamera

The total assembled results of a single image capture from the image sensor.

```
public sealed class TotalCaptureResult : CaptureResult
```

Inheritance

object ← CameraMetadata ← CaptureResult ← TotalCaptureResult

Inherited Members

CaptureResult.GetCameraId() , CaptureResult.GetFrameNumber() , CaptureResult.GetRequest() , CaptureResult.GetSequenceId() , CameraMetadata.Native , CameraMetadata.KeyProviderClass , CameraMetadata.GetConstant(string) , CameraMetadata.GetKeys() , CameraMetadata.TryGet<T>(string, out T) , CameraMetadata.TryGet<T>(CameraMetadata.Key, out T) , CameraMetadata.Dispose()

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

TotalCaptureResult(AndroidJavaObject)

```
public TotalCaptureResult(AndroidJavaObject native)
```

Parameters

`native` AndroidJavaObject

Methods

GetPartialResults()

Returns the list of partial results that compose this total result.

```
public CaptureResult[] GetPartialResults()
```

Returns

CaptureResult[]

Class YUVConverter

Namespace: Uralstech.UXR.QuestCamera

Converts raw camera capture session frames to Unity-supported RGBA.

```
public sealed class YUVConverter
```

Inheritance

object ← YUVConverter

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

YUVConverter(Resolution, GraphicsFormat)

Creates a new converter with the shader and kernel described in the scene instance of QuestCameraManager.

```
public YUVConverter(Resolution resolution, GraphicsFormat textureFormat = null)
```

Parameters

`resolution` Resolution

`textureFormat` GraphicsFormat

If not specified, uses equivalent of RenderTextureFormat.ARGB32.

YUVConverter(Resolution, ComputeShaderKernel, GraphicsFormat)

```
public YUVConverter(Resolution resolution, ComputeShaderKernel kernel,
GraphicsFormat textureFormat = null)
```

Parameters

`resolution` Resolution

`kernel` [ComputeShaderKernel]

`textureFormat` GraphicsFormat

If not specified, uses equivalent of RenderTextureFormat.ARGB32.

Fields

Texture

The output texture with converted frames.

```
public readonly RenderTexture Texture
```

Field Value

RenderTexture

Properties

CaptureTimestamp

The capture timestamp of the last processed frame.

```
public long CaptureTimestamp { get; }
```

Property Value

long

HasNewFrame

true if a capture was processed this frame; false otherwise.

```
public bool HasNewFrame { get; }
```

Property Value

bool

ShaderKernel

The shader used for conversion.

```
public ComputeShaderKernel ShaderKernel { get; set; }
```

Property Value

[ComputeShaderKernel]

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

OnFrameReady(nint, long, nint, nint, long, int, int, int, long)

Processes a frame received from the native capture session.

```
public void OnFrameReady(nint yBuffer, long yBufferSize, nint uBuffer, nint
vBuffer, long uvBufferSize, int yRowStride, int uvRowStride, int uvPixelStride,
long timestamp)
```

Parameters

`yBuffer` nint

The pointer to this frame's Y (luminance) data.

`yBufferSize` long

The size of the Y buffer in bytes.

`uBuffer` nint

The pointer to this frame's U (color) data.

`vBuffer` nint

The pointer to this frame's V (color) data.

`uvBufferSize` long

The size of the U and V buffers in bytes.

`yRowStride` int

The size of each row of the image in `yBuffer` in bytes.

`uvRowStride` int

The size of each row of the image in `uBuffer` and `vBuffer` in bytes.

`uvPixelStride` int

The size of a pixel in a row of the image in `uBuffer` and `vBuffer` in bytes.

`timestamp` long

The timestamp the frame was captured at in nanoseconds.

Events

OnFrameProcessed

Callback for when a frame has been dispatched for conversion, with the frame texture and capture timestamp.

```
public event Action<RenderTexture, long>? OnFrameProcessed
```

Event Type

Action<RenderTexture, long>

Namespace Uralstech.UXR.QuestCamera.GLES

Classes

GLESAPI

Exposes the native GLES Texture Conversion API.

GLESCaptureSession

Manages a camera capture session with a repeating request, but supports both repeating and on-demand conversion.

GLESCaptureSession.Proxy

Java proxy to handle native callbacks.

Structs

RenderJobDisposeData

Data for Dispose.

RenderJobRunData

Data for Run.

RenderJobSetupData

Data for Setup.

Enums

RenderJobEvent

IDs of all Render Job events.

Delegates

RenderJobDisposeData.Callback

Callback for when the job is disposed or the process fails.

RenderJobRunData.Callback

Callback for when the job finishes rendering or the process fails.

RenderJobSetupData.Callback

Callback for when the job is setup or the process fails.

Class GLESAPI

Namespace: Uralstech.UXR.QuestCamera.GLES

Exposes the native GLES Texture Conversion API.

```
public static class GLESAPI
```

Inheritance

object ← GLESAPI

Fields

DisposeCallbacksRegistry

Registry of job disposal callbacks. This is a single-call registry, i.e. the entry is removed after the callback occurs.

```
public static readonly ConcurrentDictionary<uint, RenderJobDisposeData.Callback>
DisposeCallbacksRegistry
```

Field Value

ConcurrentDictionary<uint, RenderJobDisposeData.Callback>

RenderJobDisposeCallbackPtr

Static marshalled pointer to OnRenderJobDispose(bool, uint).

```
public static readonly nint RenderJobDisposeCallbackPtr
```

Field Value

nint

RenderJobRunCallbackPtr

Static marshalled pointer to OnRenderJobRun(long, uint).

```
public static readonly nint RenderJobRunCallbackPtr
```

Field Value

nint

RenderJobSetupCallbackPtr

Static marshalled pointer to OnRenderJobSetup(uint, uint).

```
public static readonly nint RenderJobSetupCallbackPtr
```

Field Value

nint

RunCallbacksRegistry

Registry of job run callbacks.

```
public static readonly ConcurrentDictionary<uint, RenderJobRunData.Callback>
RunCallbacksRegistry
```

Field Value

ConcurrentDictionary<uint, RenderJobRunData.Callback>

SetupCallbacksRegistry

Registry of job setup callbacks. This is a single-call registry, i.e. the entry is removed after the callback occurs.

```
public static readonly ConcurrentDictionary<uint, RenderJobSetupData.Callback>
SetupCallbacksRegistry
```

Field Value

ConcurrentDictionary<uint, [RenderJobSetupData](#).[Callback](#)>

Methods

OnRenderJobDispose(bool, uint)

Callback for when the job is disposed or the process fails.

```
public static void OnRenderJobDispose(bool result, uint renderTextureId)
```

Parameters

`result` bool

Whether the operation was a success or not.

`renderTextureId` uint

[RenderTextureId](#), for lookup.

OnRenderJobRun(long, uint)

Callback for when the job finishes rendering or the process fails.

```
public static void OnRenderJobRun(long timestamp, uint renderTextureId)
```

Parameters

`timestamp` long

The timestamp returned by the SurfaceTexture, or -1 if the operation failed.

`renderTextureId` uint

RenderTextureId, for lookup.

OnRenderJobSetup(uint, uint)

Callback for when the job is setup or the process fails.

```
public static void OnRenderJobSetup(uint nativeTextureId, uint renderTextureId)
```

Parameters

`nativeTextureId` uint

`renderTextureId` uint

RenderTextureId, for lookup.

getGLESManageConverterJobEvent()

Returns a pointer to the native render job management function.

```
public static extern nint getGLESManageConverterJobEvent()
```

Returns

nint

Class GLESCaptureSession

Namespace: Uralstech.UXR.QuestCamera.GLES

Manages a camera capture session with a repeating request, but supports both repeating and on-demand conversion.

```
public sealed class GLESCaptureSession :
CaptureSessionBase<GLESCaptureSession.Proxy>
```

Inheritance

object ← StatefulResource ← CaptureSessionBase<GLESCaptureSession.Proxy> ←
GLESCaptureSession

Inherited Members

CaptureSessionBase<GLESCaptureSession.Proxy>.OnSessionConfigured ,
CaptureSessionBase<GLESCaptureSession.Proxy>.OnSessionConfigurationFailed ,
CaptureSessionBase<GLESCaptureSession.Proxy>.OnSessionRequestSet ,
CaptureSessionBase<GLESCaptureSession.Proxy>.OnSessionRequestSetWithId ,
CaptureSessionBase<GLESCaptureSession.Proxy>.OnSessionRequestFailed ,
CaptureSessionBase<GLESCaptureSession.Proxy>.OnSessionClosed ,
CaptureSessionBase<GLESCaptureSession.Proxy>.NativeProxy ,
CaptureSessionBase<GLESCaptureSession.Proxy>.TryAbortCaptures(out
CaptureSessionBase<GLESCaptureSession.Proxy>.ErrorCode) ,
StatefulResource.State , StatefulResource.WaitForInitialization() ,
StatefulResource.WaitForInitialization(TimeSpan, Action, WaitTimeoutMode) ,
StatefulResource.WaitForInitializationAsync(CancellationToken)

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

Texture conversion is done through a native OpenGL-ES plugin.

Constructors

GLESCaptureSession(Resolution, GraphicsFormat)

```
public GLESCaptureSession(Resolution resolution, GraphicsFormat textureFormat
= null)
```

Parameters

`resolution` Resolution

`textureFormat` GraphicsFormat

If not specified, uses equivalent of RenderTextureFormat.ARGB32.

Exceptions

NotSupportedException

Thrown if this constructor is invoked in an environment where OpenGL ES 3 is unavailable.

Fields

Texture

The output texture with converted frames.

```
public readonly Texture2D Texture
```

Field Value

Texture2D

Properties

CaptureTimestamp

The capture timestamp of the last processed frame.

```
public long CaptureTimestamp { get; }
```

Property Value

long

HasNewFrame

true if a capture was processed this frame; false otherwise.

```
public bool HasNewFrame { get; }
```

Property Value

bool

Methods

DisposeAsync()

Closes the session (if not already closed) and releases native resources.

```
public override ValueTask DisposeAsync()
```

Returns

ValueTask

ProcessSingleFrameAsync(CancellationToken)

Processes a single frame and returns the result.

```
public ValueTask<(long, Texture2D)> ProcessSingleFrameAsync(CancellationToken token
= default)
```

Parameters

`token` CancellationToken

Returns

ValueTask<(long, Texture2D)>

Capture timestamp and updated texture. Timestamp will be -1 if the capture could not be processed.

Exceptions

InvalidOperationException

Thrown if continuous processing is active.

ObjectDisposedException

TimeoutException

SetupJobAsync()

Registers the texture and creates a job in the native C++ manager.

```
public ValueTask<uint> SetupJobAsync()
```

Returns

ValueTask<uint>

The ID of the source texture created for the job, which the capture session will render to, or 0 if the operation failed.

StartContinuousProcessing(int)

Starts continuous frame processing.

```
public void StartContinuousProcessing(int maxFramerate = 60)
```

Parameters

`maxFramerate` int

The maximum rate at which frames will be processed by the GLES pipeline.

Exceptions

InvalidOperationException

Thrown if continuous processing is already active.

ObjectDisposedException

Events

OnFrameProcessed

Callback for when a frame has been processed, with the frame texture and capture timestamp.

```
public event Action<Texture2D, long>? OnFrameProcessed
```

Event Type

Action<Texture2D, long>

Remarks

The image at this point has not *actually* finished processing, but all GPU commands to do so have been executed.

Class GLESCaptureSession.Proxy

Namespace: Uralstech.UXR.QuestCamera.GLES

Java proxy to handle native callbacks.

```
public sealed class GLESCaptureSession.Proxy :
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase
```

Inheritance

object ← CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase ←
GLESCaptureSession.Proxy

Inherited Members

CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.ModifyRequestBuilder ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.ShouldRegisterCaptureEvents ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnConfigured ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnConfigureFailed ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnRequestSet ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnRequestSetWithId ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnRequestFailed ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnCaptureCompleted ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnCaptureFailed ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceCompleted ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceAborted ,
CaptureSessionBase<GLESCaptureSession.Proxy>.ProxyBase.OnClosed

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

All event callbacks will be on a Java thread, and are performance sensitive.

Struct RenderJobDisposeData

Namespace: Uralstech.UXR.QuestCamera.GLES

Data for Dispose.

```
public readonly struct RenderJobDisposeData
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

RenderJobDisposeData(uint, nint)

```
public RenderJobDisposeData(uint renderTextureId, nint onDone)
```

Parameters

`renderTextureId` uint

`onDone` nint

Fields

OnDone

Method with signature of RenderJobDisposeData.Callback.

```
public readonly nint OnDone
```

Field Value

nint

RenderTextureId

The Job's ID and render target.

```
public readonly uint RenderTextureId
```

Field Value

uint

Delegate RenderJobDisposeData.Callback

Namespace: Uralstech.UXR.QuestCamera.GLES

Callback for when the job is disposed or the process fails.

```
public delegate void RenderJobDisposeData.Callback(bool result,
uint renderTextureId)
```

Parameters

`result` bool

Whether the operation was a success or not.

`renderTextureId` uint

RenderTextureId, for lookup.

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Enum RenderJobEvent

Namespace: Uralstech.UXR.QuestCamera.GLES

IDs of all Render Job events.

```
public enum RenderJobEvent
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Fields

`Dispose = 2`

Disposes the native texture and cleans up the registered job.

`Run = 3`

Runs a job.

`Setup = 1`

Sets up a native texture to convert from, and creates a registerable job for the capture session.

Struct RenderJobRunData

Namespace: Uralstech.UXR.QuestCamera.GLES

Data for Run.

```
public readonly struct RenderJobRunData
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

RenderJobRunData(uint, nint)

```
public RenderJobRunData(uint renderTextureId, nint onDone)
```

Parameters

`renderTextureId` uint

`onDone` nint

Fields

OnDone

Method with signature of RenderJobRunData.Callback.

```
public readonly nint OnDone
```

Field Value

nint

RenderTextureId

The Job's ID and render target.

```
public readonly uint RenderTextureId
```

Field Value

uint

Delegate RenderJobRunData.Callback

Namespace: Uralstech.UXR.QuestCamera.GLES

Callback for when the job finishes rendering or the process fails.

```
public delegate void RenderJobRunData.Callback(long timestamp, uint renderTextureId)
```

Parameters

`timestamp` long

The timestamp returned by the SurfaceTexture, or -1 if the operation failed.

`renderTextureId` uint

RenderTextureId, for lookup.

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Struct RenderJobSetupData

Namespace: Uralstech.UXR.QuestCamera.GLES

Data for Setup.

```
public readonly struct RenderJobSetupData
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

RenderJobSetupData(uint, int, int, nint)

```
public RenderJobSetupData(uint renderTextureId, int width, int height, nint onDone)
```

Parameters

`renderTextureId` uint

`width` int

`height` int

`onDone` nint

Fields

Height

The height of RenderTextureId.

```
public readonly int Height
```

Field Value

int

OnDone

Method with signature of RenderJobSetupData.Callback.

```
public readonly nint OnDone
```

Field Value

nint

RenderTextureId

The GLES ID of the texture which will be the Job's ID and render target.

```
public readonly uint RenderTextureId
```

Field Value

uint

Width

The width of RenderTextureId.

```
public readonly int Width
```

Field Value

int

Delegate RenderJobSetupData.Callback

Namespace: Uralstech.UXR.QuestCamera.GLES

Callback for when the job is setup or the process fails.

```
public delegate void RenderJobSetupData.Callback(uint nativeTexture,
uint renderTextureId)
```

Parameters

`nativeTexture` uint

The created texture, or 0 if the operation failed.

`renderTextureId` uint

RenderTextureId, for lookup.

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Namespace Uralstech.UXR.QuestCamera.Vulkan

Classes

VulkanAPI

Exposes the native Vulkan Texture Conversion API.

VulkanContinuousCaptureSession

Manages a camera capture session with a continuous/repeating request.

VulkanContinuousCaptureSession.Proxy

Java proxy to handle native callbacks.

VulkanOnDemandCaptureSession

Manages a continuous capture session with user-invoked texture conversion.

Structs

ManagedRenderCallback

A managed callback for a VulkanAPI render event.

RenderData

Data for a VulkanAPI render event.

Delegates

RenderData.Callback

Callback for when the event completes.

VulkanContinuousCaptureSession.OnFrameReadyCallback

Signals the Vulkan plugin that a frame is ready for conversion.

Struct ManagedRenderCallback

Namespace: Uralstech.UXR.QuestCamera.Vulkan

A managed callback for a VulkanAPI render event.

```
public readonly struct ManagedRenderCallback
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

ManagedRenderCallback(nint, long, Action<bool, long>)

```
public ManagedRenderCallback(nint renderDataMemory, long timestampNs, Action<bool,
long> onDone)
```

Parameters

`renderDataMemory` nint

`timestampNs` long

`onDone` Action<bool, long>

Fields

OnDone

The managed callback (bool Success, long TimestampNs).

```
public readonly Action<bool, long> OnDone
```

Field Value

Action<bool, long>

Remarks

This will be invoked from the render thread, do not call Unity APIs from it.

RenderDataMemory

The native memory allocated for event data.

```
public readonly nint RenderDataMemory
```

Field Value

nint

TimestampNs

The timestamp of the frame being rendered, in nanoseconds.

```
public readonly long TimestampNs
```

Field Value

long

Struct RenderData

Namespace: Uralstech.UXR.QuestCamera.Vulkan

Data for a VulkanAPI render event.

```
public readonly struct RenderData
```

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Constructors

RenderData(nint, int, ulong, nint, nint)

```
public RenderData(nint sourceHardwareBuffer, int sourceHardwareBufferDataSpace,
ulong sourceHardwareBufferId, nint destinationImage, nint onDone)
```

Parameters

`sourceHardwareBuffer` nint

`sourceHardwareBufferDataSpace` int

`sourceHardwareBufferId` ulong

`destinationImage` nint

`onDone` nint

Fields

Align

Alignment of this struct (UnsafeUtility.AlignOf<T>()).

```
public static readonly int Align
```

Field Value

int

DestinationImage

The target Unity-created VkImage to render to.

```
public readonly nint DestinationImage
```

Field Value

nint

OnDone

Method with signature of [RenderData.Callback].

```
public readonly nint OnDone
```

Field Value

nint

Size

Size of this struct (UnsafeUtility.SizeOf<T>()).

```
public static readonly int Size
```

Field Value

int

SourceHardwareBuffer

The source AHardwareBuffer to render from.

```
public readonly nint SourceHardwareBuffer
```

Field Value

nint

SourceHardwareBufferDataSpace

The data space of SourceHardwareBuffer.

```
public readonly int SourceHardwareBufferDataSpace
```

Field Value

int

SourceHardwareBufferId

The global unique ID of SourceHardwareBuffer.

```
public readonly ulong SourceHardwareBufferId
```

Field Value

ulong

Delegate RenderData.Callback

Namespace: Uralstech.UXR.QuestCamera.Vulkan

Callback for when the event completes.

```
public delegate void RenderData.Callback(byte success, ulong hardwareBufferId)
```

Parameters

`success` byte

Was the event executed successfully?

`hardwareBufferId` ulong

SourceHardwareBufferId, for lookup.

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Class VulkanAPI

Namespace: Uralstech.UXR.QuestCamera.Vulkan

Exposes the native Vulkan Texture Conversion API.

```
public static class VulkanAPI
```

Inheritance

object ← VulkanAPI

Fields

RenderCallbackPtr

Static marshalled pointer to OnRenderDone(byte, ulong).

```
public static readonly nint RenderCallbackPtr
```

Field Value

nint

RenderCallbacksRegistry

Registry of render callbacks.

```
public static readonly ConcurrentDictionary<long, ManagedRenderCallback>
RenderCallbacksRegistry
```

Field Value

ConcurrentDictionary<long, ManagedRenderCallback>

Methods

AllocateRenderData(ref RenderData)

Allocates native memory of a RenderData struct to use with OnRenderDone(byte, ulong)/FreeRenderData(in nint).

```
public static nint AllocateRenderData(ref RenderData renderData)
```

Parameters

`renderData` RenderData

The data to allocate.

Returns

nint

A pointer to the allocated memory.

Remarks

Allocations are made with UnsafeUtility and Allocator.TempJob, matching de-allocation in OnRenderDone(byte, ulong) and FreeRenderData(in nint).

FreeRenderData(in nint)

Frees the memory of a RenderData struct allocated using AllocateRenderData(ref RenderData).

```
public static void FreeRenderData(in nint renderDataPtr)
```

Parameters

`renderDataPtr` nint

The memory to free.

OnRenderDone(byte, ulong)

Callback for when the event completes.

```
public static void OnRenderDone(byte success, ulong hardwareBufferId)
```

Parameters

`success` byte

Was the event executed successfully?

`hardwareBufferId` ulong

SourceHardwareBufferId, for lookup.

getVulkanRenderEvent()

Returns a pointer to the native rendering function.

```
public static extern nint getVulkanRenderEvent()
```

Returns

nint

releaseHardwareBuffer(nint)

Releases an acquired AHardwareBuffer object.

```
public static extern void releaseHardwareBuffer(nint acquiredBufferPtr)
```

Parameters

`acquiredBufferPtr` nint

The buffer to release.

Class VulkanContinuousCaptureSession

Namespace: Uralstech.UXR.QuestCamera.Vulkan

Manages a camera capture session with a continuous/repeating request.

```
public class VulkanContinuousCaptureSession :
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>
```

Inheritance

object ← StatefulResource ←
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy> ←
VulkanContinuousCaptureSession

Derived

VulkanOnDemandCaptureSession

Inherited Members

CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionConfigured ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionConfigurationFailed
,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionRequestSet ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionRequestSetWithId ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionRequestFailed ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionClosed ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.NativeProxy ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>._native ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>._disposed ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.MakeProxy<T>(out T) ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.TryAbortCaptures(out
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ErrorCode) ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.CloseWork() ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ThrowIfDisposed() ,
StatefulResource.State , StatefulResource.WaitForInitialization() ,
StatefulResource.WaitForInitialization(TimeSpan, Action, WaitTimeoutMode) ,
StatefulResource.WaitForInitializationAsync(CancellationToken)

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

Texture conversion is done through a native Vulkan plugin.

Constructors

VulkanContinuousCaptureSession(Resolution, GraphicsFormat)

```
public VulkanContinuousCaptureSession(Resolution resolution, GraphicsFormat
textureFormat = null)
```

Parameters

`resolution` Resolution

`textureFormat` GraphicsFormat

If not specified, uses equivalent of RenderTextureFormat.ARGB32.

Exceptions

NotSupportedException

Thrown if this constructor is invoked in an environment where Vulkan is unavailable or if the current runtime's Android API Level is lower than 33 (Android 13).

Fields

Texture

The output texture with converted frames.

```
public readonly RenderTexture Texture
```

Field Value

RenderTexture

Properties

CaptureTimestamp

The capture timestamp of the last processed frame.

```
public long CaptureTimestamp { get; }
```

Property Value

long

HasNewFrame

[true](#) if a capture was processed this frame; [false](#) otherwise.

```
public bool HasNewFrame { get; }
```

Property Value

bool

Methods

DispatchFrameConversionAsync(nint, int, long, long)

```
protected Task DispatchFrameConversionAsync(nint acquiredBufferPtr, int
bufferDataSpace, long bufferId, long timestampNs)
```

Parameters

`acquiredBufferPtr` nint

`bufferDataSpace` int

`bufferId` long

`timestampNs` long

Returns

Task

DisposeAsync()

Closes the session (if not already closed) and releases native resources.

```
public override ValueTask DisposeAsync()
```

Returns

ValueTask

OnFrameReadyNative(nint, int, long, long)

```
protected virtual void OnFrameReadyNative(nint acquiredBufferPtr, int bufferDataSpace,
long bufferId, long timestampNs)
```

Parameters

`acquiredBufferPtr` nint

`bufferDataSpace` int

`bufferId` long

`timestampNs` long

Events

OnFrameProcessed

Callback for when a frame has been processed, with the frame texture and capture timestamp.

```
public event Action<RenderTexture, long>? OnFrameProcessed
```

Event Type

Action<RenderTexture, long>

Remarks

The image at this point has not *actually* finished processing, but all GPU commands to do so have been enqueued.

Delegate VulkanContinuousCaptureSession.OnFrameReadyCallback

Namespace: Uralstech.UXR.QuestCamera.Vulkan

Signals the Vulkan plugin that a frame is ready for conversion.

```
public delegate void VulkanContinuousCaptureSession.OnFrameReadyCallback(nint
acquiredBufferPtr, int bufferDataSpace, long bufferId, long timestampNs)
```

Parameters

`acquiredBufferPtr` nint

The HardwareBuffer associated with the frame.

`bufferDataSpace` int

The DataSpace of the buffer.

`bufferId` long

The global unique ID (AHardwareBuffer_getId) of the buffer.

`timestampNs` long

The timestamp the frame was captured at in nanoseconds.

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

AHardwareBuffer_acquire has already been called on `acquiredBufferPtr`, and this method **must** release it either through native plugin render events or releaseHardwareBuffer(nint).

Class VulkanContinuousCaptureSession.Proxy

Namespace: Uralstech.UXR.QuestCamera.Vulkan

Java proxy to handle native callbacks.

```
public sealed class VulkanContinuousCaptureSession.Proxy :
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase
```

Inheritance

object ← CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase ←
VulkanContinuousCaptureSession.Proxy

Inherited Members

CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.ModifyRequestBuilder ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.ShouldRegisterCaptureEvents ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnConfigured ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnConfigureFailed ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnRequestSet ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnRequestSetWithId ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnRequestFailed ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureCompleted ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureFailed ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceCompleted ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnCaptureSequenceAborted ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ProxyBase.OnClosed

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

All event callbacks will be on a Java thread, and are performance sensitive.

Constructors

Proxy()

```
public Proxy()
```

Methods

Invoke(string, nint)

```
public override nint Invoke(string methodName, nint javaArgs)
```

Parameters

`methodName` string

`javaArgs` nint

Returns

nint

Events

OnFrameReady

Signals the Vulkan plugin that a frame is ready for conversion.

```
public event VulkanContinuousCaptureSession.OnFrameReadyCallback? OnFrameReady
```

Event Type

VulkanContinuousCaptureSession.OnFrameReadyCallback

Remarks

AHardwareBuffer_acquire has already been called on `acquiredBufferPtr`, and this method **must** release it either through native plugin render events or releaseHardwareBuffer(nint).

Class VulkanOnDemandCaptureSession

Namespace: Uralstech.UXR.QuestCamera.Vulkan

Manages a continuous capture session with user-invoked texture conversion.

```
public sealed class VulkanOnDemandCaptureSession : VulkanContinuousCaptureSession
```

Inheritance

object ← StatefulResource ←
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy> ←
VulkanContinuousCaptureSession ← VulkanOnDemandCaptureSession

Inherited Members

VulkanContinuousCaptureSession.OnFrameProcessed ,
VulkanContinuousCaptureSession.HasNewFrame , VulkanContinuousCaptureSession.Texture ,
VulkanContinuousCaptureSession.CaptureTimestamp ,
VulkanContinuousCaptureSession.DisposeAsync() ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionConfigured ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionConfigurationFailed
,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionRequestSet ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionRequestSetWithId ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionRequestFailed ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.OnSessionClosed ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.NativeProxy ,
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.TryAbortCaptures(out
CaptureSessionBase<VulkanContinuousCaptureSession.Proxy>.ErrorCode) ,
StatefulResource.State , StatefulResource.WaitForInitialization() ,
StatefulResource.WaitForInitialization(TimeSpan, Action, WaitTimeoutMode) ,
StatefulResource.WaitForInitializationAsync(CancellationToken)

Extension Methods

JNIExtensions.ToJava(object, Type) , JNIExtensions.ToJava<T>(T)

Remarks

Texture conversion is done through a native Vulkan plugin.

Constructors

VulkanOnDemandCaptureSession(Resolution, GraphicsFormat)

```
public VulkanOnDemandCaptureSession(Resolution resolution, GraphicsFormat
textureFormat = null)
```

Parameters

`resolution` Resolution

`textureFormat` GraphicsFormat

If not specified, uses equivalent of RenderTextureFormat.ARGB32.

Exceptions

NotSupportedException

Thrown if this constructor is invoked in an environment where Vulkan is unavailable or if the current runtime's Android API Level is lower than 33 (Android 13).

Methods

OnFrameReadyNative(nint, int, long, long)

```
protected override void OnFrameReadyNative(nint acquiredBufferPtr, int
bufferDataSpace, long bufferId, long timestampNs)
```

Parameters

`acquiredBufferPtr` nint

`bufferDataSpace` int

`bufferId` long

`timestampNs` long

ProcessSingleFrameAsync(CancellationToken)

Requests and processes a single frame and returns the result.

```
public ValueTask<(long, RenderTexture)> ProcessSingleFrameAsync(CancellationToken
token = default)
```

Parameters

`token` CancellationToken

Returns

ValueTask<(long, RenderTexture)>

Capture timestamp and updated texture.

Remarks

The image when returned has not *actually* finished processing, but all GPU commands to do so have been enqueued.

Exceptions

ObjectDisposedException

RequestFrame()

Requests that the next frame be processed (converted).

```
public void RequestFrame()
```

Exceptions

ObjectDisposedException